Appendix: Methodology for the Conjoint Analysis Question

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June 3, 2026

Contents

1 Overview

This appendix describes in full detail the construction of the conjoint analysis question included in the Global Justice questionnaire. The conjoint presents respondents with pairs of stylised societal scenarios and asks them to choose between them. Each scenario is characterised by eight underlying parameters (Section 3) and displayed through seven features (Section ??). The income distributions underlying the features are derived from Bothe et al. (2026) and Chancel et al. (2026); temperature projections rely on the emission pathways from Chancel et al. (2026). All monetary quantities are expressed in EUR PPP 2025 per adult per year unless otherwise noted.

2 Scenario nomenclature

A scenario is characterised by four structural dimensions:

Hours (or class). The average weekly working hours converge to a 2100 target corresponding to one of: `S15k`, `S45k`, `S` (= `S60k`), `M`, `P` (from shortest to longest; see Table 1).

Scope. The redistribution configuration: `C` = Convergence (both national SN and global GIT); `G` = Global only; `N` = National only; `I` = Inequality (neither).

Sectoral change. `1` = no sectoral change (stable public services, current food system); `2` = structural change (expanded public services, reduced beef and flights).

Decarbonization. `SD` = slow; `ID` = intermediate; `FD` = fast.

By convention, the scope is embedded in the hours label (e.g. `SC45k2_FD` for 30 h/wk, Convergence, structural change, fast decarbonization). The default benchmark is `SC2_FD` (35 h, Convergence, structural change, fast decarbonization).

Table 1: Mapping between working hours and 2100 GDP per capita (k€/adult).

Weekly hours	Hours label	With GIT (k€)	Without GIT (k€)
25	S15k	15	12.5
30	S45k	45	37.5
35	S	60	50
40	M	79	66
45	P	120	100

3 Eight underlying parameters

Each respondent profile is defined by one parameter representing the respondent themselves and seven policy-relevant parameters:

1. **Individual income**: the respondent's household monthly cash income and whether they live as a couple. The annual individual income is $y_0 = \text{HH\_monthly} \times 12/k$, where $k = 2$ if couple, $k = 1$ otherwise.

2. **Decarbonization**: Slow (SD), Intermediate (ID), Fast (FD), corresponding to IEA CPS, STEPS and NZE pathways respectively. The income cost at 2035 is 0 % (SD), 2 % (ID) or 4 % (FD).

3. **Working hours**: target weekly hours per scenario class (25, 30, 35, 40, 45 hours).

4. **National redistribution**: current level, or Sustainable National redistribution (SN) as modelled in Bothe et al. (2026).

5. **Global redistribution**: no global tax (current), or the Global Income Tax (GIT) of Bothe et al. (2026).

6. **Public services**: stable (20 % of GNI) or increased (+3 percentage points of GNI, financed by a flat-rate tax).

7. **Beef and flights**: no change; −60 % beef; −50 % air travel; both.

The full design yields $5 \times 2 \times 2 \times 3 \times 2 \times 4 = 480$ unique scenarios.

4 Income distribution data sources

4.1 National secondary income from Bothe et al.

The Bothe et al. simulation provides, for each country c, gpercentile p and year t:

- $\text{sdiinc}_{c,p,t}$: the income share of bracket p in country c;

- $\text{nni}_{c,t}$: national net income per adult (EUR PPP 2025);
- $\text{diff}_{c,t}$: a normalising scalar;
- $\text{ypt}_{c,p,t}$: the global income tax (GIT) paid;
- div_t: the uniform global dividend received by every adult.

The resulting per-adult full income is:

$$\text{income}_{c,p,t} = \underbrace{\frac{\text{sdiinc}_{c,p,t} \cdot \text{nni}_{c,t}}{\text{diff}_{c,t}}}_{\text{yp\_recomp: national secondary income, pre-GIT}} - \text{ypt}_{c,p,t} + \text{div}_t. \tag{1}$$

yp_recomp is the national secondary income *after* all national taxes and transfers but *before* the global income tax. All four redistribution scopes are therefore defined as:

$$\begin{aligned}
\text{SC}_{c,p,t} &= \text{income}_{c,p,t} = \text{yp_recomp} - \text{ypt} + \text{div}, && (2)\\
\text{SN}_{c,p,t} &= (\text{SC}_{c,p,t} - \text{GR}_t)\left(G_c^{SI_0}/G_c^{SC}\right), && (3)\\
\text{SI}_{c,p,t} &= \tfrac{1}{2}\,\text{PI}_{c,p,t} \quad \text{(in 2100; see Section 6)}, && (4)\\
\text{SG}_{c,p,t} &= \left(G_c^{SC}/G_c^{SI_0}\right)\text{SI}_{c,p,t}. && (5)
\end{aligned}$$

where $G_c^{SC}, G_c^{SI_0}$ are the gp-weighted mean incomes of the SC and SI ($= \frac{1}{2}$PI) distributions and GR $=$ div $-$ ypt is the net global transfer (so SC $-$ GR $=$ yp_recomp). The N scope rescales the nationally-redistributed income to the no-global GDP level G^{SI_0}; the G scope rescales the SI shape to the converged level G^{SC}.

Point of attention: Bothe et al. provide a single SC simulation. The SC1 and SC2 variants (sectoral change off/on) and all decarbonization paces (SD/ID/FD) share the same income distribution. The differences between them are modelled *explicitly* by us through (a) an income-cost multiplier for decarbonization (Section 5.3) and (b) a flat-rate PS tax for structural change (Section 5.4).

4.2 Cash income for Italy 2025

The conjoint uses *gross cash income*: primary income at market prices, net of all taxes (including VAT) and net of imputed rents and retained earnings, but *gross* of capital consumption. This corresponds to what a respondent would recognise as their disposable income plus employer social contributions, comparable across scenarios.

Starting from Fisher-Gethin & Chancel 2023 (Fisher-Gethin and Chancel, 2023) data for Italy:

$$\text{cash}_p = \left(a_p^{\text{pre}} - \tau_p^{\text{dir}} - \tau_p^{\text{soc}} - \tau_p^{\text{ind}} + g_p^{\text{soc}}\right)\text{NNI} - r_e\,\kappa_p\,\text{NNI} - r_h\,H_p\,\text{NNI} + \kappa_p\,\text{CFC} - \text{ypt}_p, \tag{6}$$

where:

- a_p^{pre} = normalised pretax income share at percentile p;
- τ_p^{dir}, τ_p^{soc}, τ_p^{ind} = direct taxes (PIT + wealth + CIT), social contributions, and indirect taxes (VAT), each normalised by mean pretax income;
- g_p^{soc} = government cash transfers;
- $r_e = 0.50 \times \bar{m}$ = retained-earnings deduction rate ($\bar{m}$ = WID corporate profit retention ratio for Italy);
- $r_h = 0.035$ = imputed-rent yield on net housing wealth (Garbinti et al., 2021);
- H_p = normalised housing-wealth allocation at percentile p (interpolated from the 2014 French wealth-composition data of Garbinti et al. 2021);
- κ_p = capital-income share at percentile p;
- CFC = capital consumption per adult (so as to gross up from NNI to GNI for the capital share);
- ypt_p = GIT tax already paid in 2025.

We enforce monotonicity on the bottom-99 % of the distribution by sorting values in ascending order, mirroring the procedure in the main distributional series. All values are expressed in EUR PPP 2025.

Point of attention: τ^{ind} (VAT) is subtracted because the Fisher-Gethin pretax income is at market prices (inclusive of indirect taxes). Subtracting VAT converts to factor-cost NNI, to which cash incomes sum. Omitting this step would overstate cash income.

5 IT distributions at 2035

5.1 Base distribution: IT35

Let IT35_p denote the gross cash income distribution for Italy at 2035 under `SC1_SD` (slow decarbonization, no structural change). It serves as the universal base from which all 2035 IT scenario distributions are derived. We compute it by rescaling the 2035 Bothe income distribution to cash units using the 2025 individual cash-to-income ratio:

$$\text{IT35}_p = \text{income}_{IT,p,2035} \times \frac{\text{cash}_p}{\text{income}_{IT,p,2025}}. \tag{7}$$

This per-percentile ratio preserves the distributional shape while anchoring the level to the respondent-observable 2025 cash income.

5.2 GR35: the 2035 net global redistribution transfer

The per-percentile net global redistribution transfer in 2035 is

$$\text{cash\_GR35}_p = \left(\text{div}_{2035} - \text{ypt}_{p,2035}\right) \times \frac{\text{cash}_p}{\text{income}_{IT,p,2025}}. \tag{8}$$

cash_GR35_p is positive for low-income individuals (net GIT receivers) and negative for high-income individuals (net GIT contributors).

> **Point of attention:** Monotonicity is *not* enforced on `cash_GR35`. Unlike income distributions, which must be weakly increasing, the net GIT transfer is mechanically decreasing with income (higher earners pay more than they receive). Sorting it would scramble the distributional signal.

5.3 Decarbonization income factor

We model the income cost of decarbonization as a flat multiplicative factor applied to all 2035 distributions:

$$\delta = \begin{cases} 1.00 & \text{(SD)} \\ 0.98 & \text{(ID)} \\ 0.96 & \text{(FD)} \end{cases} \tag{9}$$

This captures a $-4\,\%$ income penalty for fast decarbonization relative to slow, consistent with the income-side impact of accelerated investment in the energy transition.

> **Point of attention:** The factor δ is applied *on top of* the Bothe income distribution. Bothe et al. report the same income distribution regardless of the decarbonization pace (the SD, ID and FD variants differ only in emissions and temperature). The δ multiplier is therefore an explicit parametric cost not already embedded in the simulation.

5.4 Public-services flat-rate tax

When public services are expanded (`food=2`), the additional expenditure is financed by a flat-rate tax at rate

$$\tau_{\text{PS}} = \frac{(s_{2035} - s_{2025}) \times \bar{g}_{2035}}{\bar{\text{cash}}_{2035}}, \tag{10}$$

where s_t = share of educ+health+PS spending in GNE (from Chancel sheet `G5s`), $\bar{g}_{2035}$ = per-adult GNE at 2035 (`G0p`), and $\bar{\text{cash}}_{2035}$ = average of IT35. For Italy: $\tau_{\text{PS}} \approx 4.9\,\%$.

5.5 Scenario-specific 2035 IT distributions

All 2035 IT columns are derived from IT35 using the following formulas, where $\delta_{\text{FD}} = 0.96$ and $\psi = 1 - \tau_{\text{PS}}$:

$$\begin{aligned}
\texttt{IT35} &= \mathrm{IT35}_p && \text{(SC1_SD: base)} && (11)\\
\texttt{SCmat} &= \mathrm{IT35}_p \cdot \delta_{\mathrm{FD}} && \text{(SC1_FD)} && (12)\\
\texttt{SC} &= \mathrm{IT35}_p \cdot \delta_{\mathrm{FD}} \cdot \psi && \text{(SC2_FD)} && (13)\\
\texttt{PC} &= 1.15\,\mathrm{IT35}_p \cdot \delta_{\mathrm{FD}} && \text{(PC1_FD)} && (14)\\
\texttt{PI} &= 1.4\,\mathrm{IT25}_p \cdot \delta_{\mathrm{FD}} && \text{(PI1_FD)} && (15)\\
\texttt{SC45k} &= (G_{2025}/G_{2035})\,\mathrm{IT35}_p \cdot \delta_{\mathrm{FD}} \cdot \psi && && (16)\\
\texttt{SC15k} &= 0.9\,(G_{2025}/G_{2035})\,\mathrm{IT35}_p \cdot \delta_{\mathrm{FD}} \cdot \psi && && (17)\\
\texttt{SI} &= (G^{PI}_{2035}/G_{2025})\,(h^{SC}_{2035}/h^{PI}_{2035})\,\mathrm{IT25}_p \cdot \delta_{\mathrm{FD}} \cdot \psi && && (18)\\
\texttt{SN} &= (\mathrm{IT35}_p - \mathrm{cash_GR35}_p)\,(G^{SI_0}/\overline{\mathrm{SC}}) \cdot \delta_{\mathrm{FD}} \cdot \psi && && (19)\\
\texttt{SG} &= (\overline{\mathrm{SC}}/G^{SI_0})\,\mathrm{SI0}_p \cdot \delta_{\mathrm{FD}} \cdot \psi && && (20)\\
\texttt{MC} &= \Phi\,\mathrm{IT35}_p \cdot \delta_{\mathrm{FD}} \cdot \psi && && (21)
\end{aligned}$$

where G_t denotes Italy's per-capita GDP at year t (Chancel `A0p`; G^{PI}_{2035} is the PI per-capita GDP, `A0pi`); $h^{SC}_{2035}, h^{PI}_{2035}$ are per-capita economic labour hours under SC/PI (Chancel `E0h`/`E0k`); $\mathrm{SI0}_p = (G^{PI}_{2035}/G_{2025})(h^{SC}/h^{PI})\,\mathrm{IT25}_p$ is the SI base, with $\overline{\mathrm{SC}}, G^{SI_0}$ the gp-weighted means of the SC and SI bases; and Φ is the MC income scale factor defined in Section 5.6.

Point of attention: `PI` uses IT25 (2025 cash income) as its base, not IT35, because the PI scenario does not undergo the SC productivity and redistribution transition between 2025 and 2035. It grows from the current distribution by a flat GDP-per-capita factor of 1.4. By contrast, `SC` and its variants start from IT35, which already embeds the Bothe SC income structure at 2035.

Point of attention: `SI` uses IT25 as its base, scaled to the no-global GDP level: the PI per-capita GDP adjusted for SC's shorter working hours, $(G^{PI}_{2035}/G_{2025})(h^{SC}/h^{PI})$. It preserves the current within-country shape (no redistribution). `SN` and `SG` then sit at the same no-global level G^{SI_0} (SN) and the converged level $\overline{\mathrm{SC}}$ (SG): `SN` rescales the SC-redistributed income (IT35 − GR35) to G^{SI_0}, and `SG` rescales the SI base to $\overline{\mathrm{SC}}$.

5.6 MC scenario: moderate convergence

The MC scenario holds per-capita working hours constant at their 2025 level while following the SC long-run productivity path (converging to 125 €/h by 2100). The 2035 income scale relative to IT35 is:

$$\Phi = \frac{\bar{g}^{10}}{\omega_{2035} \cdot \phi_{2035}}, \qquad (22)$$

where $\bar{g} = (125/\phi_{2025})^{1/75}$ is the constant annual growth rate in hourly productivity under SC (from 2025 level ϕ_{2025} to 125 €/h in 2100), $\phi_{2035} = \phi_{2035}/\phi_{2025}$ is the actual SC productivity growth 2025–2035 (from Chancel sheet `F0a`), and $\omega_{2035} = h_{2035}/h_{2025}$ is the SC per-capita-hours change 2025–2035 (sheet `E0h`).

Point of attention: In the SC scenario, productivity growth is front-loaded (peaking around 2080 before declining), so ϕ_{2035} can exceed the steady-state rate $\bar{g}^{10}$. In that case $\Phi < 1$, meaning MC income at 2035 is *lower* than SC income — a plausible result because SC extracts peak productivity via its 35 h week while MC works 40 h at a lower per-hour frontier.

6 Income distributions at 2100

6.1 Income concept and rescaling

At 2100, within-country income comparisons across scenarios and between countries are best made using "full income" (national secondary income inclusive of all public services), rather than cash income. However, to keep the 2100 income comparable with the 2035 cash income entered by the respondent (and shown in the IT35/SC columns), we rescale by a *single scalar*:

$$r_{\mathrm{IT}} = \frac{\overline{\mathrm{cash}}_{IT,2025}}{\overline{\mathrm{income}}_{IT,2025}}, \tag{23}$$

where both averages are taken over the 127-bracket Italy 2025 distribution weighted by bracket widths. In the data $r_{\mathrm{IT}} \approx 0.69$.

This ratio is applied uniformly across all gpercentiles and all countries. The resulting 2100 distributions therefore represent cash-income-equivalent magnitudes, stripped of the within-country variation in the cash-to-income ratio. The consistency check is: $r_{\mathrm{IT}} \times \bar{y}_{IT,2100} \approx r_{\mathrm{IT}} \times \bar{y}_{World,2100}$ in the SC scenario (since SC achieves near-complete global convergence by 2100), which holds in the data.

Point of attention: The decision to use a *scalar* r_{IT} for 2100 (rather than the per-percentile ratio used for 2035) is deliberate. For 2035, the per-percentile ratio is needed to preserve distributional comparability with the cash income entered by the respondent. For 2100, cross-country and cross-scenario comparability is the priority, which requires a *uniform* rescaling so that the shape of the 2100 world distribution reflects true inequality rather than the Italy-specific cash-to-income heteroscedasticity.

6.2 IT 2100 distributions

For Italy at 2100, in raw full-income units (before r_{IT}):

$$\texttt{SC}_{IT,2100} = \mathrm{income}_{IT,p,2100} \quad \text{(Bothe SC: GIT+dividend)} \tag{24}$$

$$\texttt{SN}_{IT,2100} = (\mathrm{income}_{IT,p,2100} - \mathrm{div}_{2100})\,(G_{IT}^{SI_0}/G_{IT}^{SC}) \quad ((\mathrm{SC}{-}\mathrm{GR}) \times G^{SI_0}/G^{SC}) \tag{25}$$

$$\texttt{PI}_{IT,2100} = \mathrm{yp\_recomp}_{IT,p,2025} \times g_{IT}^{\mathrm{PI}} \quad (g^{\mathrm{PI}} = A0\mathrm{pi}_{2100}/A0\mathrm{pi}_{2025}) \tag{26}$$

$$\texttt{SI}_{IT,2100} = \tfrac{1}{2}\,\texttt{PI}_{IT,2100} \tag{27}$$

$$\texttt{SG}_{IT,2100} = (G_{IT}^{SC}/G_{IT}^{SI_0}) \times \texttt{SI}_{IT,2100} \quad (G^{SC}, G^{SI_0} = \text{mean of SC, SI}) \tag{28}$$

For hours variants and PC, apply the coefs from Table 2 to the `SC` distribution:

$$\mathrm{X}_{IT,2100} = c_X \times \texttt{SC}_{IT,2100}, \qquad c_X \in \{2.0,\ 1.0,\ 0.75,\ 0.25,\ 630/480\} \text{ for PC, SC, SC45k, SC15k, MC.} \tag{29}$$

Table 2: 2100 income coefs relative to SC. Applied to both IT and World distributions.

Scenario Hours	SC15k 25h	SC45k 30h	SC (=SCmat) 35h	MC 40h	PC 45h
Coef	0.25	0.75	1.00	630/480	2.00

Point of attention: `PI` base uses yp_recomp$_{2025}$ (pre-GIT national secondary income), not income$_{2025}$ (which includes GIT transfers). The PI scenario has no global income tax; using income$_{2025}$ would inflate the PI distribution for net GIT receivers and depress it for net contributors.

Point of attention: `SCmat` and `SC` have identical income distributions at 2100. Structural change affects emissions and temperature (and therefore 2035 incomes via δ) but not 2100 full incomes: by 2100, both modalities converge to the same equilibrium income level in Bothe et al.'s simulation.

6.3 World 2100 distributions

The world distribution is built by pooling all (country, gpercentile) cells, weighting each cell by $w_{c,p} = \Delta_p \times \mathrm{Pop}_c$, sorting by income value, and binning into 100 equal-population global percentiles. Cells with non-positive or non-finite income are dropped. Gaps in the resulting 100-grid are filled by linear interpolation.

Populations used follow each scenario's GDP level: Pop^{SC} (Chancel sheet `Z0a`, ≈ 9.41 B in 2100, Convergence) for SC, SG, SC45k, SC15k, MC and PC; and Pop^{PI} (`Z0b`, ≈ 10.18 B, Inequality) for PI, SI and SN.

Point of attention: PC inherits the SC world distribution scaled by 2.0, using SC population. A more rigorous treatment would use the UN-medium population for PC (as Chancel et al. 2026 recommend), but the implied distributional error is small since country-level growth differences matter more than aggregate population.

6.4 Output format for ineq_2100

The file `ineq_2100.csv` contains 21 income brackets: p_0p_5, p_5p_{10}, $\ldots$, $p_{90}p_{95}$, $p_{95}p_{99}$, $p_{99}p_{100}$. Each IT bracket is the gp-width-weighted mean over the 127-grid; each World bracket is the unweighted mean over the corresponding 100-grid bins. All values are rounded to the nearest integer EUR.

7 Temperature at 2100

7.1 Base temperature

Temperature depends on GDP (which encodes working hours and global redistribution), decarbonization pace, and sectoral change. For scenarios present in Chancel's `Emission_Output` workbook, we use T_{final} from `chancel_temp2100_completed.csv`. For scenarios not modelled by Chancel (SI, SN, SG, MC and their variants), we predict temperature using a GDP-based no-emissions regression estimated on the 24 Chancel-observed scenarios:

$$T = \beta_0 + \beta_1 G + \beta_2 \mathbf{1}_{\text{ID}} + \beta_3 \mathbf{1}_{\text{SD}} + \beta_4 G\, \mathbf{1}_{\text{ID}} + \beta_5 G\, \mathbf{1}_{\text{SD}} + \beta_6 c\, \mathbf{1}_{\text{FD}} + \beta_7 c\, \mathbf{1}_{\text{ID}} + \beta_8 c\, \mathbf{1}_{\text{SD}} + \varepsilon, \quad (30)$$

where G = total 2100 GDP (T€), $\mathbf{1}_d$ = indicator for decarbonization pace d, $c = \mathbf{1}(\text{structural change} = 1)$, and ε is the residual. Adjusted $R^2 \approx 0.997$.

The total 2100 GDP is computed as $G = g_{\text{pc}} \times P$, where g_{pc} is GDP per capita from Table 1 and P is the 2100 population (9.41 B for SC family, 10.18 B for PI/no-GIT at 45h).

7.2 Temperature adjustments for beef, flights and public services

Following Chancel et al. (2026), once a "canonical" modality is fixed, additive corrections are applied:

- **Beef reduction**: $-0.24\,°\text{C}$.
- **Flight reduction**: $-0.155\,°\text{C}$.
- **Expanded public services** (relative to no expansion): $\pm\, 0.0004875 \times (67.25 + 2.06\, g_{\text{pc}})\,°\text{C}$, where g_{pc} is in k€/adult. This formula is derived from the regres-

sion of fossil emissions on GDP per capita across SC scenarios (Section 17 of the `questionnaire_legacy.R` script).

The canonical modality is chosen as follows (Chancel et al., 2026, method_questionnaire.md):

SD and ID: Fix the change according to the public-services choice (stable $\to$ 1, increased $\to$ 2). Adjust for beef and flights relative to the canonical co-variation with structural change.

FD: Fix the change according to the beef choice (no reduction $\to$ 1, beef reduced $\to$ 2). Land-use effects dominate at fast decarbonization. Adjust for flights and public services (the latter already captured via the sectoral-change dummy in equation 30).

Point of attention: Temperature is rounded to the nearest 0.1 °C. In the no-emissions regression the maximum absolute deviation from Chancel observed temperatures is 0.1 °C (by construction: the model was selected to guarantee this on the training set). For extrapolated scenarios (SI/SN/SG/MC) the true error is unknown but bounded by the regression's generalization uncertainty.

8 Own income at 2035

1. Compute the respondent's annual individual income: $y_0 = y_{\mathrm{HH}} \times 12/k$.
2. Find the respondent's gpercentile p^* in the IT25 distribution by linear interpolation on the 127-bracket cumulative distribution.
3. Retrieve their 2035 income $\hat{y}_{2035}$ by interpolating the appropriate scenario column of `ineq_IT_2035.csv` at p^*.

The scenario column is determined by the user's working hours, redistribution scope, decarbonization and public-services choices (after applying δ and ψ corrections to the exported base column).

For hours levels not equal to 35h and redistribution scope other than GIT+national, the base column (GIT+national at the chosen hours level) is scaled by the ratio $\bar{y}_{\mathrm{scope}}/\bar{y}_{\mathrm{SC}}$, where both averages are the weighted mean of the corresponding 35h columns in `ineq_IT_2035.csv`.

Point of attention: The IT25 distribution may not be monotone over the full 127 gpercentile range due to imputation noise in the top-1 % brackets. The percentile search is therefore applied only to the 99-bracket bottom distribution; above gpercentile 99, the last bracket's income is returned.

9 2100 income bar charts

The *national incomes* chart shows the IT 2100 distribution in 21 brackets and the *global incomes* chart shows both the IT and the World 2100 distributions. All values are cash-income-equivalent (scaled by r_{IT}).

For hours variants not equal to 35h, the ineq_2100 distribution for the given redistribution scope is scaled by:

$$c_{\mathrm{hours}} = \frac{\bar{W}_{\mathrm{hours}}}{\bar{W}_{\mathrm{SC}}}, \tag{31}$$

where $\bar{W}$ denotes the bracket-weighted world average from `ineq_2100.csv`. This ratio equals the hours coef from Table 2 for the GIT+national scope, and generalises to other scopes by the same proportionality.

> **Point of attention:** The bar charts display *average* income within each bracket. The top bracket ($p_{99}p_{100}$) typically shows a very steep increase: in the SC scenario this bracket is compressed by redistribution; in the PI/SI scenarios it can be an order of magnitude above the median. When displaying these charts in the questionnaire, consider truncating or log-scaling the y-axis to avoid the top bracket dominating the visual.

10 Consistency checks and known limitations

1. **SN vs SI**: both sit at the no-global GDP level G^{SI_0} (same average) but differ in shape: SN starts from the nationally-redistributed income (IT35−GR35, i.e. SC's compressed shape) and SI from the 2025 distribution (IT25). The wedge between them at intermediate percentiles is the effect of national redistribution alone.

2. **SG vs SC**: SG rescales the SI shape to the converged SC level ($\mathtt{SG} = (G^{SC}/G^{SI_0})\,\mathtt{SI}$), so it shares SC's average but keeps the no-national-redistribution (2025) shape. The G scope (global redistribution only) thus lifts incomes to the converged level while leaving within-country inequality at its current shape.

3. **IT_SC = World_SC at 2100**: Under SC, countries converge to approximately the same per-capita income by 2100, so Italy's 2100 distribution and the world's 2100 distribution should have similar averages. This is confirmed in the data ($\bar{y}_{IT,SC} \approx \bar{y}_{World,SC}$ after applying r_{IT}).

4. **PI base for 2100**: The PI base uses yp_recomp$_{2025}$ (pre-GIT national secondary income) rather than income$_{2025}$ (which includes the 2025 GIT dividend). Using the post-dividend income would overstate PI incomes for net GIT receivers and understate them for net contributors.

5. **No PI2 scenarios**: The PI scenario does not model sectoral change in Chancel's data (sectoral_change = 0 for all PI variants, regardless of the public-services choice). This is a limitation: PI+expanded PS is computed using structural change = 0, so the temperature does not reflect any emissions benefit from sectoral reallocation.

6. **MC temperature**: MC is not in Chancel's emission output workbook. Its temperature is predicted by the no-emissions regression, which may not generalise well to the MC GDP level (741 T€), which lies between SC (565 T€) and PC (1129 T€) but has a distinct productivity path.

11 Data files

`ineq_IT_2035.csv` 127 rows (gpercentile) × 14 columns (IT25, cash_GR35, IT35, SCmat, SC, PC, PI, SC45k, SC15k, SI, SN, SG, MC). Income in EUR PPP 2025/adult/yr.

`ineq_2100.csv` 21 rows (brackets) × 21 columns (bracket label + IT and World for each of the 10 scenarios). Income in EUR PPP 2025/adult/yr × r_{IT}.

`temp100.csv` 54 rows: 30 Chancel scenarios + 24 new rows for SI/SN/SG/MC at all decarb × food combinations. Columns: scenario, gdp (T€), sectoral_change, decarb, temp_final (°C).

`conjoint_constants.csv` 54 key-value rows covering: r_{IT}, τ_{PS}, decarbonization factors, temperature adjustment coefficients, GDP per capita by hours × GIT, population projections, no-emissions regression coefficients, and average incomes from both distribution files.

12 Reproducibility

All distributions are generated by `code_simulator/questionnaire.R`, which is self-contained (reads raw Bothe and Chancel data files plus `chancel_temp2100_completed.csv`) and produces the four output files above. The conjoint computation at display time is handled by `code_simulator/js/conjoint.js`, which reads the CSV files via `fetch()` and requires a web server.

References

Bothe, O., Chancel, L., and co-authors (2026). Replication package for Global Justice Report Working Paper 2026-11. *WID.world* Working Paper. URL: https://wid.world/document/replication-package-working-paper-2026-11/

Chancel, L., and co-authors (2026). Global Justice Report 2026. *World Inequality Lab*. Data: https://wseed.world/data.html

Fisher-Gethin, A. and Chancel, L. (2023). Distributional National Accounts: Methods and Estimates for 50 Countries. *Quarterly Journal of Economics* (forthcoming).

Garbinti, B., Goupille-Lebret, J. and Piketty, T. (2021). Accounting for Wealth Inequality Dynamics: Methods, Estimates and Simulations for France. *Journal of the European Economic Association* 19(1), 620–663.